

Static Gesture Classification — Run Report

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Data paths

Training data root	/home/jestin/ThesisRepo/ML/NewTrainingData/Static
External test root	/home/jestin/ThesisRepo/ML/NewTestData/Henry/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz
Normalisation	standard
Feature mode	stats → feature dim 1408
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=680 · test=85

Augmentation (training only)

Enabled	yes
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
SVM (RBF)	0.9971	0.0088	1.0000
SVM (Linear)	1.0000	0.0000	1.0000
Random Forest	1.0000	0.0000	1.0000
Logistic Regression	0.9971	0.0088	1.0000
Voting Soft (SVM+RF+KNN+LogReg)	0.9971	0.0088	1.0000
Stacking (SVM+RF+KNN+LogReg)	0.9971	0.0088	1.0000
KNN	0.9809	0.0175	0.9882
LDA	0.9338	0.0289	0.5176

Per-class hold-out F1

Class	SVM (RBF)	SVM (Linear)	Random Forest	Logistic Regression	Voting (soft)	Stacking	KNN	LDA	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.500	13
Double_Nothing	1.000	1.000	1.000	1.000	1.000	1.000	0.957	0.118	12
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.667	12
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.621	12
Double_Phone	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.400	12
Double_Pistol	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.600	12
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	1.000	0.960	0.526	12
macro avg	1.000	1.000	1.000	1.000	1.000	1.000	0.988	0.490	85

External test (NewTestData)

Algorithm	External test acc
Stacking (SVM+RF+KNN+LogReg)	0.8000
Voting Soft (SVM+RF+KNN+LogReg)	0.7714
SVM (Linear)	0.7429
Logistic Regression	0.7429
Random Forest	0.6571
KNN	0.6286
SVM (RBF)	0.2000
LDA	0.1714

Per-class external accuracy

Class	Stacking	Voting (soft)	SVM (Linear)	Logistic Regressi on	Random Forest	KNN	SVM (RBF)	LDA	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	0.800	1.000	0.000	0.000	5
Double_Nothing	1.000	0.800	0.600	0.600	0.800	0.800	1.000	0.400	5
Double_Okay	0.800	0.800	1.000	0.600	1.000	0.800	0.000	0.400	5
Double_Open_Palm	1.000	1.000	0.600	1.000	1.000	0.800	0.000	0.200	5
Double_Phone	0.600	0.600	1.000	1.000	0.000	0.000	0.200	0.000	5
Double_Pistol	0.200	0.200	0.000	0.000	0.000	0.400	0.000	0.200	5
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	0.600	0.200	0.000	5